PLEASE WRITE YOUR ROLL NUMBER **Roll No.** 22B0763

CODE **A**



CE 343 **S2** Water Resources Engineering
25 April 2025 Friday 1330-1630 (SLOT 3)
Time: 3 h **weightage: 40 %**

1. **Write your name, roll number and signature on the (i) question paper and data sheet (p.3 of the question paper) (ii) main answer book.**
2. THERE ARE SIX QUESTIONS ON 3 PAGES INCLUDING THE DATA SHEET
3. **THERE IS NO PARTIAL MARKING!**
4. Start each answer on a fresh page and Write the Question number clearly and **circle it!**
5. **Show one set of steps and calculations (to 3 decimal places) clearly in the answer book.**
6. **There are no marks for just the answers!! Also, NO marks will be awarded if step wise calculations are NOT shown.**
7. **Make suitable assumptions where necessary (no marks for WRONG assumptions!)**
8. **The terms have the usual notations DO NOT ASK THE TA's ! (minus marks for asking!!)**
9. **Remember to attach the completed table to the answer book!!**

1. A 30-cm diameter well completely penetrates an unconfined aquifer of saturated depth 40 m. After a long period of pumping at a steady rate of 1500 litres per minute, the drawdown in two observation wells 25 m and 75 m from the pumping well were observed to be 3.5 m and 2.0 m respectively. **15**

(a) Draw a neat sketch showing the radial flow to the well with the data,
(b) Determine the transmissibility of the aquifer (in **m^2/s**), and
(c) Also determine the drawdown (in **m**) in the main pumped well.

2. There are two mutually exclusive water storage projects - Project A and Project B for a city. Project A provides an initial investment large enough to meet the demands for water for 40 years with a single dam, and project B uses investment to make the dam in two stages to meet the same demand. **15**

Analyse the two water storage projects using the **present worth method** and determine which project should be selected.

Show the figures, steps and calculations clearly

The data for the two projects is given in the following table:

	Project A	**Project B**
	Costs in ₹Lakhs	Costs in ₹Lakhs
Construction cost	350.0	250.0 1st stage 300.0 2nd stage
Operations and maintenance	1.5 per year for 40 years	1.0 per year for 1st 20 years 2.0 per year for 2nd 20 years
Annual benefit	30.0	30.0
Economic life	40 years	40 years (20 years for each stage)
Period of analysis	40 years	40 years
Discount rate	5 percent	5 percent

3. (i) Design a regime channel in alluvium for a discharge of 25.0 m^3/s and silt factor 1.1, using Lacey's Regime Theory. Take z= 0.5H:1V, Take the freeboard as 0.75 m. Indicate the bed slope as 1 in X **15**

(ii) Sketch the channel section details with all dimensions.

4. Determine the design head (Clearly show the stepwise calculations for all three iterations) iterations) for an Ogee spillway with the following data: **15**

i. Height of spillway crest above river bed = 100m
ii. Design discharge = 10,000 cumecs
iii. Number of spans = 6
iv. Clear distance between piers = 15 m
v. Thickness of pier = 3m
vi. Slope of d/s face of the overflow section = 0.8: 1

Neglect the end contractions and assume the value of C_d as 2.20.

Assume, pier contraction coefficient K_p = 0.02 and

abutment contraction coefficient K_a = 0.2

For downstream profile, take K = 2 and n = 1.85

Assume any other data if required.

Make suitable assumptions – list them clearly.

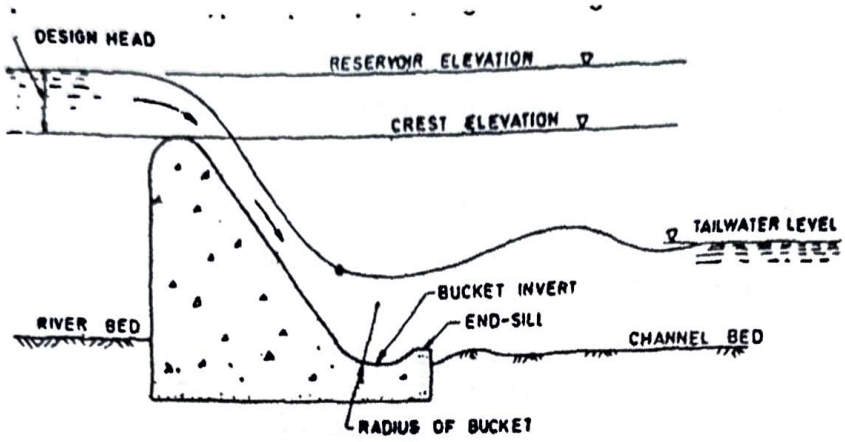


FIG. 1 OGEE SPILLWAY

5.	During a low water week, a river stream gets an average daily flow of 35 cumecs with daily fluctuations requiring a pondage capacity of 16% of the daily flow. A hydroelectric plant is to be located on this river, which will operate for 5 days a week, 24 hours a day, but will supply power at varying rates, such that the daily load factor is 60 %, corresponding to which the pondage required is 0.2 times the mean flow to the turbines. On Saturdays and Sundays, all flow is ponded for use on the remaining 5 days. If the effective head on the turbine, when the pond is full is desired to be 26 m, and the maximum allowable fluctuation in the pond level is 1.2 m, determine: (i) The surface area (in Ha) of the pond to satisfy all the operating conditions, and (ii) The weekly output at the switchboard in kWh. Assume the turbine efficiency at 82% and the generator efficiency as 90%	15
6.	A reservoir is proposed to be constructed to command an area equal to 1,00,000 hectares. The crops to be grown, their monthly water requirements, and the area under irrigation of each of these crops as a percentage of command is shown in Table 6. Determine (show stepwise calculations): (i) The annual storage required for the reservoir, assuming canal losses as 25% of head discharge, and reservoir evaporation and dead storage losses as 20% of gross capacity, and (ii) The channel capacity at the outlet point. Assume that the peak demand is 1.25 times the average demand. **SHOW ONE SET OF STEPWISE CALCULATIONS FOR THE MONTH OF NOVEMBER FOR EACH CROP** and complete the rest of the data in the attached Table.	15

PLEASE WRITE YOUR ROLL NUMBER

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Roll No. 2200768

Month	days	FIR (Field irrigation requirement) in cm						Month	Water required (ha.m)						Total vol. water reqd all crops ha.m	Discharge ha.m/day
		Paddy	Ground nut	Maize	Cotton	Sugar cane	Chillies		Paddy	Ground nut	Maize	Cotton	Sugar cane	Chillies		
% area under irrign		25	5	5	15	15	5	Area (ha)								
Jun	30	19				26		Jun	4750	–	–	–	3900	–	8650	288.333
Jul	31	7				8		Jul	1750	–	–	–	1200	–	2950	95.161
Aug	31	7				7		Aug	1750	–	–	–	1050	–	2800	90.323
Sep	30	8				6	2	Sep	2000	–	–	–	900	100	3000	100
Oct	31	30		3	4	35	24	Oct	7500	–	150	600	5250	1200	14700	474.194
Nov	30	21	6	15	8	42	23	Nov	5250	300	750	1200	6300	1150	14950	498.333
Dec	31		16	24	21	18	27	Dec	–	800	1200	3150	2700	1350	9200	296.774
Jan	31		22	21	23	22	11	Jan	–	1100	1050	3450	3300	550	9450	304.839
Feb	28		14		14	25		Feb	–	700	–	2100	3750	–	6550	233.929
Mar	31					37		Mar	–	–	–	–	5550	–	5550	179.032
Apr	30					41		Apr	–	–	–	–	6150	–	6150	205
May	31					50		May	–	–	–	–	7500	–	7500	241.935
Given that Total Area (ha)			100000						Gross Storage Requirement			(in M m^3)	1524.167 Hm^3	✓		
									Channel Capacity			(in m^3/s)	96.128 m^3/s	10		

Aaclishree Srivastava.

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उत्तर पुस्तिका/ Answer Book-16

22B0763.
रोल नं./**Roll No.**

CE343.
पाठ्यक्रम नाम/**Course Name**

शाखा/प्रभाग/**Branch/Div.** शैक्षणिक बैच /**Tutorial Batch**

अनुभाग/**Section**

प्र.सं/Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	योग/Total
आरंभिक पृष्ठ Starting Page No.													
अंक/Marks	15	13	15	15	02	12							72

प्र.सं/Q. No.	13	14	15	16	17	18	19	20	21	22	23	24	योग/Total
आरंभिक पृष्ठ Starting Page No.													
अंक/Marks													

परीक्षा में उपस्थित होने वाले अभ्यर्थियों से घोषणा पत्र	**Declaration from the Candidate appearing in the Examination**
1. मैं, इस परीक्षा के आयोजन संबंधी पाठ्यक्रम अनुदेशक द्वारा दिए गए सभी अनुदेशों का पालन करूँगा / करूँगी।	1. I will abide by all instructions given by the course instructor regarding the conduct of this examination.
2. मैं, संस्थान के नियम तथा विनियमों में दिए गए शैक्षिक कदाचार संबंधी नियम तथा विनियमों से अवगत हूँ।	2. I am aware of the rules and regulations regarding academic misconduct as given in rules and regulations of the institute.
3. मैं, इस परीक्षा के दौरान शैक्षिक सत्यनिष्ठा के सिद्धांतों पर दृढ़ रहूँगा / रहूँगी।	3. I will adhere to the principles of academic integrity during this examination.

हस्ताक्षर/ Signature :- Aaclishree

दिनांक/ Date :- _____

② $i\% = 5\%$ (discount rate)

Present worth Method:

Single payment present worth: $(P/A, i\%, N) = \frac{1}{(1+i)^N}$

Series payment present worth: $(P/A, i\%, n) = \frac{(1+i)^n - 1}{i(1+i)^n}$

Let $P_{20} = (P/A, 5\%, 20) = \frac{1}{(1+5\%)^{20}} = 0.377$

(single payment)

$P_{40} = (P/A, 5\%, 40) = \frac{1}{(1+5\%)^{40}} = 0.142$

~~(series payment~~

(single payment)

(small p) $p_{20} = (P/A, 5\%, 20) = \frac{(1+5\%)^{20} - 1}{5\%(1+5\%)^{20}} = 12.462$

(series payment)

(small p) $p_{40} = (P/A, 5\%, 40) = \frac{(1+5\%)^{40} - 1}{5\%(1+5\%)^{40}} = 17.159$

(series payment)

Present worth of Project A:

$PWA = \sum (P/A, i, n).(B_P - C_P)$

(in ₹ Lakhs)

$= -350 - 1.5 \times p_{40} + 30 p_{40}$

$= -350 + 28.5 p_{40} = -350 + 28.5 \times 17.159$

$= 139.032$ (₹ Lakhs) $= ₹ 1,39,03,200$

$PWB = -250 - 300 P_{20} - 1 \times p_{20} - 2 \times p_{20} \times P_{20} + 30 \times p_{40}$

$= -250 - 113.1 - 12.462 - 9.396 + 514.77$

$= 129.812$ (₹ Lakhs) $= ₹ 1,29,81,200$

15

∵ PWB < PWA.

∴ ~~For~~ Project A should be selected since it has a higher present worth.

③ (i).

Lacay's Regime Theory:

$$v = \left[\frac{Q \cdot f^2}{140}\right]^{1/6} \qquad R = \frac{5}{2}\frac{v^2}{f} \qquad A = Q/v \qquad P = 4.75\sqrt{Q}$$

Given $$S = \left[\frac{f^{5/3}}{3340\, Q^{1/6}}\right]$$

Given: $Q = 25\ m^3/s$.

$f = 1.1$

$0.5H : 1V \Rightarrow z = 0.5$.

Freeboard $= 0.75\ m$.

$$v = \left[\frac{25 \times (1.1)^2}{140}\right]^{1/6} = 0.775\ m/s.$$ ✓

$$R = \frac{5}{2}\frac{v^2}{f} = \frac{5}{2} \times \frac{(0.775)^2}{1.1} = 1.365\ m.$$

$$A = Q/v = 25/0.775 = 32.258\ m^2$$ ✓

$$P = 4.75\sqrt{Q} = 23.75\ m.$$ ✓

$z = 0.5$. ✓

Designing a trapezoidal channel:

$$A = (B_w + zy)y = (B_w + y/2)y. = 32.258 \quad \text{—①}$$

$$P = B_w + 2y\sqrt{1+z^2} = B_w + \sqrt{5}\,y = 23.75 \quad \text{—②}$$ ✓

$$B_w = \frac{32.258}{y} - y/2. \quad \text{(From ①)}.$$

$$B_w = 23.75 - \sqrt{5}\,y. \quad \text{(From ②)}.$$

$$\frac{32.258}{y} - y/2 = 23.75 - \sqrt{5}\,y.$$ ✓

$$\Rightarrow 1.736\,y^2 - 23.75\,y + 32.258 = 0$$

$\Rightarrow y = 1.529\ m$ or $y = 12.152\ m$.

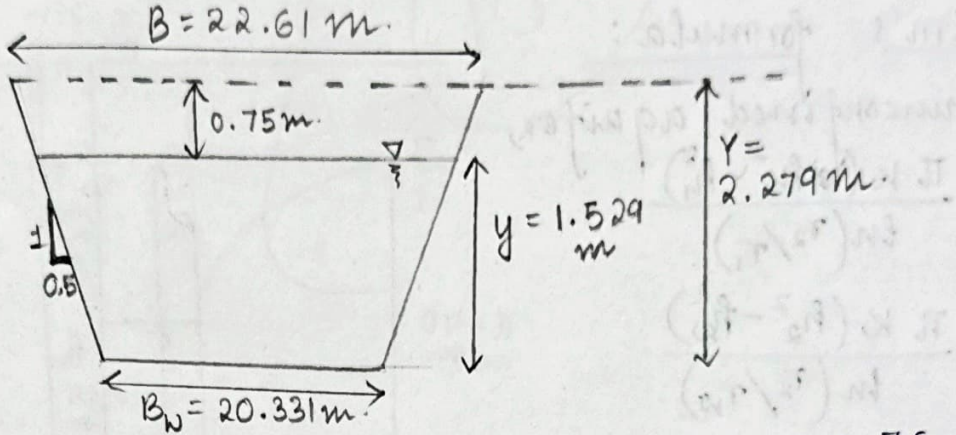
Rejected because of infeasibility.

$y = 1.529\ m$.

$\Rightarrow B_W = 23.75 - \sqrt{5}\, y = 20.331\ m$.

Bed slope $= \dfrac{(1.1)^{5/3}}{3340 \times (25)^{1/6}} = 2.05 \times 10^{-4} = 1$ in 4872.464

(ii).



$Y = y + \text{freeboard} = 1.529 + 0.75$

$= 2.279\ m$.

$B = B_W + 2Y.z = 20.331 + 2 \times 2.279 \times 0.5$.

$= 22.61\ m$.

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(1). a)

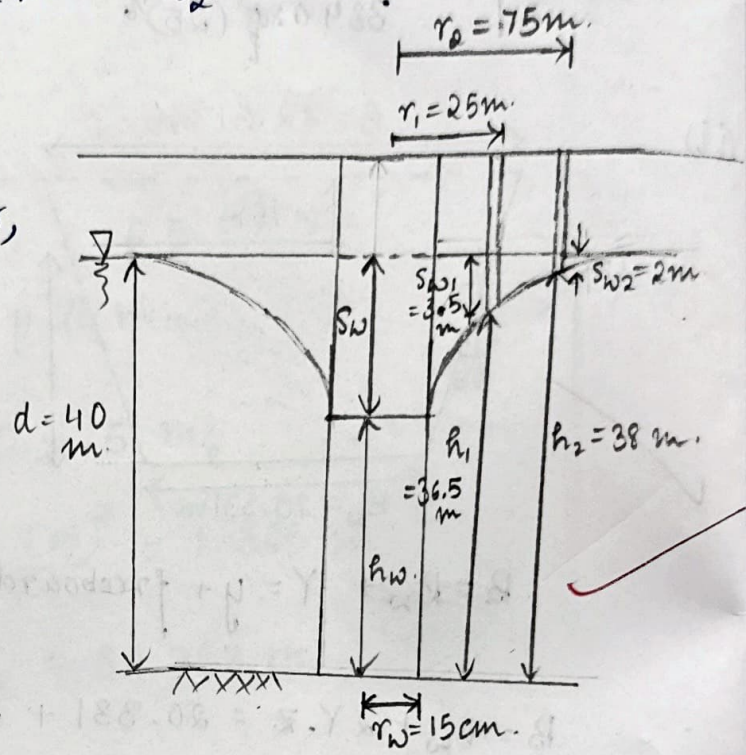
$r_w = \frac{30}{2}\,cm = 15\,cm = 0.15\,m.$

$d = 40\,m.$

$Q = 1500\ lpm = \frac{1.5}{60}\,m^3/s. = 0.025\,m^3/s.$

$r_1 = 25\,m.$ $\quad s_{w1} = 3.5\,m.$ $\quad h_1 = d - s_{w1} = 36.5\,m.$

$r_2 = 75\,m.$ $\quad s_{w2} = 2\,m.$ $\quad h_2 = d - s_{w2} = 38\,m.$



Theim's formula:

For unconfined aquifer,

$$Q = \frac{\pi K (h_2^2 - h_1^2)}{\ln(r_2/r_1)}$$

$$Q = \frac{\pi K (h_2^2 - h_w^2)}{\ln(r_2/r_w)}$$

k: coefficient of permeability.

(b). $$Q = \frac{\pi K (h_2^2 - h_1^2)}{\ln(r_2/r_1)}$$

$$\Rightarrow 0.025 = \frac{\pi \cdot K (38^2 - 36.5^2)}{\ln(75/25)} \Rightarrow K = 7.823 \times 10^{-5}\ m/s.$$

$\Rightarrow$ Transmiesablity $T = Kd = 7.823 \times 10^{-5} \times 40$

$\Rightarrow T = 3.129 \times 10^{-3}\ m^2/s.$

(c) $$Q = \frac{\pi K (h_2^2 - h_w^2)}{\ln(r_2/r_w)} \Rightarrow 0.025 = \frac{\pi \times 7.823 \times 10^{-5} (38^2 - h_w^2)}{\ln(75/0.15)}$$

$\Rightarrow 38^2 - h_w^2 = 632.165.$

$\Rightarrow h_w^2 = 811.835.$

$\Rightarrow h_w = 28.493\ m.$

$h_w = d -$

$s_w = d - h_w = 11.507\ m$

∴ Drawdown in the main pumped well is 11.507m.

10

(4).

$Q = 10000\ m^3/s.$

Number of piers N = no: of spans − 1

$\Rightarrow N = 6 - 1$

$\Rightarrow N = 5.$

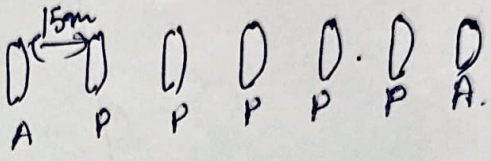
Clear distance b/w piers = ~~15m~~ $c = 15m.$

Thickness of piers = 3m.

$C_d = 2.2.$

$K_p = 0.02$

$K_a = 0.2$



Net length of overflow crest

$L' =$ no. of spans $\times c.$

$= 6 \times 15.$

$\Rightarrow L' = 90\ m.$

$Q = C_d L H_d^{3/2}$

1st iteration:

$$H = \left(\frac{Q}{C_d L'}\right)^{2/3} = \left(\frac{10000}{2.2 \times 90}\right)^{2/3} = 13.663\ m.$$

Effective length of overflow crest:

$L = L' - 2(NK_p + K_a)H.$

$= 90 - 2(5 \times 0.02 + 0.2)\,13.663.$

$= 90 - 0.6 \times 13.663$

$L = 81.802\ m.$

2nd iteration:

$$H = \left(\frac{Q}{C_d L}\right)^{2/3} = \left(\frac{10000}{2.2 \times 81.802}\right)^{2/3} = 14.562\ m.$$

$L = L' - 2(NK_p + K_a)H = 90 - 0.6 \times 14.562$

$= 81.263\ m.$

3rd iteration:

$$H_1 = \left(\frac{Q}{C_d L}\right)^{2/3} = \left(\frac{10000}{2.2 \times 81.263}\right)^{2/3} = 14.626\ m.$$

$$L = L' - 2(N K_p + K_a) H_1$$
$$= 90 - 0.6 \times 14.626.$$
$$\Rightarrow L = 81.224\ m.$$

∴ Design head for the Ogee spillway:

$$H_D = \left(\frac{Q}{C_d L}\right)^{2/3} = \left(\frac{10000}{2.2 \times 81.224}\right)^{2/3}$$

$$\Rightarrow H_D = 14.631\ m.$$

15

⑥. (ii) A = 1,00,000 ha.

Calculations for the month of November:

Water required for paddy = Area x height

= 25% x 100000 x 21

= 525000. ha.cm.

= 5250 ha.m.

Water required for Ground nut

= 5% x 100000 x 6.

= 30000 ha.cm.

= 300 ha.m.

Water required for maize = 5% x 100000 x 0.15

= 750 ha.m.

Water required for cotton = 15% x 100000 x ~~0.21~~ 0.08

= 1200 ha.m.

Water required for sugar cane = 15% x 100000 x 0.42

= ~~27~~ 6300 ha.m

" " " chillies = ~~5~~ 5% x 100000 x 0.23

= 1150 ha.m

Total ~~vol~~ vol. water required for all crops

= 5250 + 300 + 750 + 1200 + 6300 + 1150

= 14950 ha.m.

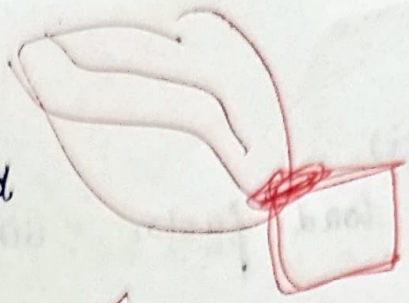
$$\text{Discharge} = \frac{\text{Total water req. monthly}}{\text{no. of days in the month}} = \frac{14950}{30}$$

= 498.333 ha.m/day.

Maximum avg. ~~monthly~~ daily ~~discharge~~ demand occurs in November.

Max. avg. ~~discharge~~ demand = $498.333 \times 10^4 / 24 \times 3600$ m^3/s.

= 57.677 m^3/s.



Peak demand = 1.25 × avg. demand
= 1.25 × 57.677
= 72.096 m^3/s.

~~100~~ Canal losses = 25%.

(100% − 25%) channel capacity = peak demand.

∴ Channel capacity at the outlet point
= $\frac{4}{3}$ × 72.096
= 96.128 m^3/s.

(i). ~~Sum of~~ Sum of total water required for all months = 8650 + 2950 + 2800 + 3000 + 14700 + 14950 + 9200 + 9450 + 6550 + 5550 + 6150 + 7500.
= 91450 ha.m.

~~75%~~ ~~75% × 80%~~
(100% − 25%) × (100% − 20%) × Gross storage requirement
= total water required = 91450 ha.m.

~~⇒ 3/4~~ ⇒ Gross storage requirement
= 91450 × $\frac{4}{3}$ × $\frac{5}{4}$.
= 152416.667 ha.m.
= 1524.167 M m^3.

5. (i).

load factor = 60% = $\frac{\text{avg. load}}{\text{peak load}}$

~~$P = 9.81 \eta$~~ $\eta_T = 82\%$. (for the 5 days)

$\eta_g = 90\%$ (for Sat and Sun)

$H = 26$ m. (effective head).

Allowable fluctuation $h = 1.2$ m.

$P = 9.81 \eta Q H$.

$Q = 35 \text{ m}^3/\text{s}$.

a). 16% pondage required for daily flow of 35 cumecs.

Volume of pondage $\leq h \times$ surface area

$\Rightarrow 16\% \times 35 \times 24 \times 3600 \leq 1.2 \times$ surface area

$\Rightarrow$ Surface area $\geq 403200 \text{ m}^2$

$= 40.32$ ha.

(b) load factor = 0.6.

$\Rightarrow$ avg load = 0.6 peak load.

Mean flow to the turbines each day = $35 \text{ m}^3/\text{s} \times 24 \times 3600$ s.

$= 3024000 \text{ m}^3$.

Pondage req. = 0.2×3024000

$= 604800 \text{ m}^3$.

~~surface area $\geq \frac{604800}{4.2}$~~ Pondage req. for 5 days $= 3024000 \text{ m}^3$

Pondage on Sat. and Sunday $= 2 \times 35 \times 24 \times 3600$

$= 6048000 \text{ m}^3$.

Avg. total output power supplied in a day

$= 9.81 \eta_T Q H \times 24 = 9.81 \times 0.82 \times 35 \times 26 \times 24$

$= 175685.328$ kWh.

Avg. load = 7320.222 kW.

Peak load = $\frac{\text{Avg. load}}{0.6} = 12200.37$ kW.

Additional load = peak load - avg. load.

= 4880.148 kW.

~~3024000~~ Additional discharge required

$= \frac{4880.148}{9.81 \eta_T H} = 23.333$ m^3/sec.

$23.333 \times$ add. time = 3024000.

$\Rightarrow$ add time = 129600 sec.

= 36 hrs.

02